Catalytic competence bounds the decay rates of a thermally formed radical pair

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(Dated: 2 November 2026)

A radical pair can transduce a weak magnetic field into a change of catalytic rate only if it carries the catalytic flux and lives long enough for the field to act. For a pair formed thermally from a closed-shell enzyme–substrate complex, returning to it only through the singlet and leaving through a spin-independent onward step, we prove that the onward yield obeys $\Phi_P \le 4k_P/(k_S+4k_P)$ for any spin Hamiltonian and any trace-preserving, unital, positivity-preserving relaxation, a bound that complete singlet–triplet mixing approaches. Detailed balance then requires $k_{\mathrm{cat}}e^{\Delta/k_BT}(4/k_S+1/k_P)\le 1$, where Δ is the free energy of the pair: spin mixing relaxes the spin-free condition only in its onward term, by the number of spin states. In a flavin–aminium spin model with anisotropic hyperfine tensors and radicals stacked 3.5–6 Å apart, a finite search finds the largest field effect among pairs that satisfy this condition for $k_{\mathrm{cat}} = 1\ \mathrm{s}^{-1}$ to fall below 0.1 % at $\Delta = 0.47$ eV; tying the exchange coupling to the back-transfer rate lowers this value only for a large, known reorganization energy. We apply the result to the single-electron-transfer pair once proposed for monoamine oxidase, using measured rates of flavin reduction and an explicit thermodynamic cycle. With the measured flavin potentials and a proxy amine potential, such a pair could not carry the observed flux, as earlier thermodynamic arguments concluded. A field effect above 0.1 % would require a low-dielectric active site, with an effective dielectric constant below about 3.1 at contact.

I. INTRODUCTION

The radical-pair mechanism (RPM) is the best-established way in which a magnetic field as weak as the geomagnetic field ($\sim 50\,\mu$T) can alter the yield of a chemical reaction.[1,2] It underlies the leading model of avian magnetoreception, in which photoexcitation of the flavin in cryptochrome creates a spin-correlated flavin–tryptophan radical pair.[3–5] Field effects on proteins that operate without light have been sought for several decades, with few reproducible results,[6] and a stopped-flow study of several flavoenzymes found no field effect on their kinetics.[7] Radical pairs have nevertheless been proposed as the origin of a range of biological field effects.[8]

The monoamine oxidases (MAO), the flavoenzymes that degrade monoamine neurotransmitters, are a natural test case. A single-electron-transfer (SET) mechanism, which would form a flavin–aminium radical pair, was proposed for them,[9,10] but Walker and Edmondson, comparing the activation free energy of flavin reduction with the free energy of electron transfer from the amine to the flavin, judged it thermodynamically and kinetically improbable; no flavin radical is observed during the reductive half-reaction,[11] and structural, kinetic and computational studies favour a polar mechanism in which the substrate α-C–H bond is cleaved without a radical intermediate.[12–15] We do not assume that MAO forms a radical pair. We ask instead a question that can be answered from the rates and the thermodynamics alone, and which extends that argument to the spin dynamics: if a thermally formed radical pair did carry the catalytic flux, which decay rates would it need, and how large a field effect could it then show?

General limits on weak-field effects are known. Electron spin relaxation is expected to limit a primary RPM effect in biochemistry to 10^{-3}–10^{-4} or less,[16] and exchange and dipolar couplings between the radicals suppress singlet–triplet interconversion unless the radicals are far apart.[17] Asymmetric reaction schemes, in which only the singlet pair recombines while both spin states react onward, relax the second limit and allow closely bound pairs to respond.[18] For a pair that recombines by non-adiabatic electron transfer, the ratio of the reactive exchange coupling to the recombination rate is fixed by the activation energy of the transfer.[19] These results constrain the spin dynamics of a pair once its rates are given. They do not say which rates a pair can have when an enzyme forms it *thermally* and turns over through it.

Here we derive that constraint. Its scope is the following: the pair (i) is formed thermally from a closed-shell state, (ii) returns to that state only through the singlet channel, (iii) carries the catalytic flux through a spin-independent onward step, and (iv) evolves under a Markovian master equation with Haberkorn reaction terms and trace-preserving, unital, positivity-preserving spin relaxation. The results do not apply to photochemically formed pairs, to mechanisms with a third radical, or to a pair whose recombination is not described by such a master equation; a pair on a dead-end branch obeys a separate occupancy bound (Sec. III). Within this scope we prove that the onward yield obeys $\Phi_P \le 4k_P/(k_S+4k_P)$ for any spin Hamiltonian and any admissible relaxation, a bound that complete singlet–triplet mixing approaches, and that detailed balance then requires $k_{\mathrm{min}}(4/k_S+1/k_P)\le 1$ with $k_{\mathrm{min}} = k_{\mathrm{cat}}e^{\Delta/k_BT}$. We combine this condition with a finite numerical search for the largest field effect at given rates in a flavin–aminium spin model in which the exchange coupling is tied to the back-transfer rate, and we apply the result to the hypothetical SET pair of MAO.

II. REACTION SCHEME AND SPIN MODEL

A. Scheme and observable

We consider the scheme

$$\mathrm{E\cdot S} \underset{k_S}{\overset{k_f}{\rightleftharpoons}} {}^1[\mathrm{Fl}^{\bullet-}\,\mathrm{RNH_2}^{\bullet+}] \xrightarrow{k_P} \mathrm{X} \longrightarrow \cdots \longrightarrow \mathrm{E_{red}\cdot P}. \quad (1)$$

Electron transfer from the closed-shell complex forms the pair in its singlet state; only the singlet pair returns by back electron transfer (rate k_S), and singlet and triplet pairs leave by a spin-independent step (rate k_P) to an intermediate X. For α-C–H deprotonation of the aminium radical, X is a neutral radical pair, which must then react further, possibly in a spin-dependent way, to give the reduced enzyme and the product imine. Aminium radical cations can also deprotonate at nitrogen: dialkylaminium radicals in water have $pK_a = 5.3$–6.8, and the diethylaminium radical deprotonates predominantly at nitrogen under the conditions studied.[20] What follows requires only that the first onward step is spin independent and is not reversed on the time scale of the pair. The reductive half-reaction of MAO B is effectively irreversible,[11] but this does not show that its first step is; we assume it. Every turnover then passes once through the first pair, so the flux through it bounds the turnover whatever the later steps are; spin-dependent later steps can only lower the flux, and field effects that arise in them are outside the scope of this work.

The yield of onward reaction per pair formed is $\Phi_P = k_P \int_0^\infty \mathrm{Tr}\,\rho\, dt$, and the magnetic field effect is $\mathrm{MFE}_P = [\Phi_P(B) - \Phi_P(0)]/\Phi_P(0)$ at $B = 50\,\mu\mathrm{T}$. Because the formation rate does not depend on the field, MFE_P is also the relative change of the flux through the pair per E·S complex. The turnover per enzyme changes less. In a cycle that passes once through the pair, the time spent in the pair, Φ_P/k_P per pair formed, lengthens the cycle, and to first order the relative change of the turnover is $\mathrm{MFE}_P/(1+x)$ with $x = k_f\Phi_P/k_P$; eq 4 below gives $x \le 4k_f/k_S = e^{-\Delta/k_BT}$, below 10^{-4} for $\Delta \ge 0.25$ eV. Where the step carried by the pair does not limit turnover, the change of k_{cat} is smaller still. MFE_P therefore bounds the relative change of enzyme activity; the absolute change of Φ_P is reported in the Supporting Information. The spin density operator of a pair born in the singlet evolves under a master equation with Haberkorn reaction terms,[19,21]

$$\begin{aligned} \dot\rho = \mathscr{L}\rho = &-2\pi i[H,\rho] - \tfrac{k_S}{2}\{P_S,\rho\} - k_P\rho + \mathscr{D}\rho, \\ &\rho(0) = P_S/N, \end{aligned} \quad (2)$$

with P_S the singlet projector, $N = \mathrm{Tr}P_S$ the nuclear-spin dimension and $\mathscr{D}$ a spin-relaxation superoperator. We assume that $\mathscr{D}$ is trace preserving, $\mathrm{Tr}\,\mathscr{D}(X) = 0$, and generates a positivity-preserving evolution, as a trace-preserving Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) generator[22,23] does, and that it is unital, $\mathscr{D}(I) = 0$, as single-electron dephasing, the reactive singlet–triplet dephasing of ref 19 and any relaxation that drives the spins towards equal populations at high temperature are. The reaction terms are derived for weak, non-adiabatic electronic coupling, with the nuclear motion that drives electron transfer fast compared with the spin dynamics.[19] Yields are obtained exactly: without relaxation from the Sylvester equation for $\int \rho\, dt$, solved by the Bartels–Stewart method,[24] and with relaxation by the same Sylvester solve inside a GMRES iteration[25] (Supporting Information).

B. Spin Hamiltonian and conventions

In frequency units,

$$H = \omega_B \hat n_B\cdot(\mathbf{S}_1+\mathbf{S}_2) + \sum_k \mathbf{S}_{e(k)}\cdot \mathsf{A}_k\cdot\mathbf{I}_k + J\,\mathbf{S}_1\cdot\mathbf{S}_2 + \mathbf{S}_1\cdot\mathsf{T}\cdot\mathbf{S}_2, \quad (3)$$

with $\omega_B = g\mu_B B/h$ and hyperfine tensors A_k. $J = E_T - E_S$ is the full singlet–triplet gap; in the convention of Efimova and Hore[17] $J = -2J_{\mathrm{EH}}$. T is the traceless dipolar tensor; for an axial tensor $\mathbf{S}_1\cdot\mathsf{T}\cdot\mathbf{S}_2 = 2D[(\mathbf{S}_1\cdot\hat n)(\mathbf{S}_2\cdot\hat n) - \mathbf{S}_1\cdot\mathbf{S}_2/3]$, so that with the field along $\hat n$ the $\mathrm{T}_\pm$ and T_0 levels are split by D ($D = -77.9$ MHz for point dipoles 1 nm apart[17]). Both conventions are verified numerically in the code.

C. Magnetic parameters

The model has four nuclei. Flavin N5 and N10 (spin 1) carry the literature isotropic couplings, 523 and 189 μT,[26] plus anisotropic spin-dipolar tensors from density-functional theory, rotated into the frame in which the dipolar coupling of the pair is computed. The aminium radical contributes the two largest couplings of the benzylaminium radical cation, 257 and 196 MHz (α-$\mathrm{CH_2}$ protons, anisotropy below 5 % of the isotropic part). All couplings are computed with B3LYP[27,28]/def2-TZVP[29] in PySCF[30] on GFN2-xTB geometries[31]; the tensors are listed in the Supporting Information. The dipolar tensor of the contact pair is the point-dipole tensor summed over the Löwdin[32] spin populations of the two radicals, as Efimova and Hore did for cryptochrome,[17] for methylamine stacked on the flavin si face at an N5–N distance of 3.5 Å, with $D = -843$ MHz and $E = -26$ MHz. We assume that substrate and flavin are stacked in the active site and scale this tensor by $s \in [0.2, 1]$, which spans stacked separations from the contact geometry to about 6 Å [$(3.5/6)^3 \approx 0.2$]; the thresholds depend on this assumption (Sec. VI). The exchange coupling is tied to k_S in Sec. IV.

D. Use of AI tools

AI tools were used extensively in this work. Claude Opus 5.5 (Anthropic), used through the Claude Code interface, wrote and ran the code for the spin dynamics, the field-effect searches and the density-functional calculations, drafted and edited the text, and searched the literature. GPT Sol 5.6 and Astra 6 (both OpenAI) were used as independent critical reviewers of the derivations, the code and the text before submission. The tools were used to implement, run and cross-check a large number of calculations in limited time, and the

reviewing tools were chosen from a different developer to reduce shared errors. The authors checked the derivations, verified the code through the self-checks and tests distributed with it (every number in the text is regenerated from the stored results), and checked every reference against its DOI record and against the statement it is cited for. The authors take full responsibility for the content of the article.

III. CATALYTIC COMPETENCE BOUNDS THE DECAY RATES

Let Δ be the free energy of the radical pair relative to E·S, summed over its four spin states, so that, with spin energies negligible against k_BT, the singlet pair lies $\Delta_S = \Delta + k_BT\ln 4$ above E·S ($k_BT\ln 4 = 0.037$ eV at 310 K) and detailed balance for the singlet channel gives $k_f = k_S\, e^{-\Delta_S/k_BT}$. The pair carries the flux $k_f\Phi_P$ per E·S complex. The turnover per enzyme is this flux times the fraction of enzyme present as E·S, which is at most one, so $k_f\Phi_P \geq k_{\mathrm{cat}}$ is necessary for the pair to carry the turnover. For the reductive half-reaction at saturating substrate, the measured limiting rate of flavin reduction is itself a rate per E·S complex.

Yield bound. For any spin Hamiltonian and any $\mathscr{D}$ that is trace preserving, unital and positivity preserving,

$$\Phi_P \leq \frac{4k_P}{k_S + 4k_P}. \tag{4}$$

The proof has three steps. (i) For $k_P > 0$ the resolvent $\mathscr{R} = (-\mathscr{L})^{-1} = \int_0^\infty e^{\mathscr{L}t}dt$ exists, and unitality gives $-\mathscr{L}(I) = k_S P_S + k_P I$, so that

$$k_S\mathscr{R}(P_S) + k_P\mathscr{R}(I) = I. \tag{5}$$

(ii) Define $\langle A,B\rangle = \mathrm{Tr}[A\mathscr{R}(B)]$ for Hermitian A, B. The trace of eq 5 after multiplication by P_S, compared with conservation of the pairs, $\Phi_S + \Phi_P = 1$ (which uses trace preservation), gives $\langle P_S, I\rangle = \langle I, P_S\rangle$. (iii) The generator is dissipative in the Hilbert–Schmidt inner product, $\mathrm{Tr}[B\mathscr{L}(B)] \leq 0$ for Hermitian B: the Hamiltonian term contributes nothing, the reaction terms contribute $-k_S\mathrm{Tr}(P_SB^2) - k_P\mathrm{Tr}B^2 \leq 0$, and $\mathrm{Tr}[B\mathscr{D}(B)] \leq 0$ because $e^{\mathscr{D}t}$ is positive, unital and trace preserving, so that Kadison's inequality[33] $(e^{\mathscr{D}t}B)^2 \preceq e^{\mathscr{D}t}(B^2)$ gives $\mathrm{Tr}(e^{\mathscr{D}t}B)^2 \leq \mathrm{Tr}B^2$. Writing $A = -\mathscr{L}(C)$ shows $\langle A,A\rangle = \mathrm{Tr}[C(-\mathscr{L})(C)] \geq k_P\mathrm{Tr}C^2 > 0$ for $A \neq 0$, so the symmetric part of $\langle\cdot,\cdot\rangle$ is positive definite and obeys the Cauchy–Schwarz inequality, $\langle P_S, I\rangle^2 \leq \langle P_S, P_S\rangle\langle I, I\rangle$. With $c = \langle I, P_S\rangle$, the traces of eq 5 and of the same identity multiplied by P_S give $\langle P_S, P_S\rangle = (N - k_Pc)/k_S$ and $\langle I, I\rangle = (4N - k_Sc)/k_P$, and the inequality becomes $c \leq 4N/(k_S + 4k_P)$; since $\Phi_P = k_Pc/N$, this is eq 4. Equivalently, a singlet-born pair spends at least a quarter of its lifetime in the singlet state. Because P_S and I are linearly independent, the inequality is strict for $k_P > 0$. Without spin dynamics $\Phi_P = k_P/(k_S + k_P)$; the bound is approached, but not attained, when singlet–triplet mixing is complete, as under fast depolarizing relaxation, where $\Phi_P(k_S + 4k_P)/4k_P = 0.999999$ (Supporting Information). Over 300 random spin systems with exchange, dipolar and hyperfine couplings and, in part, electron spin relaxation, the largest value of this ratio is 0.999996, and maximizing it over general Hermitian Hamiltonians reaches one without exceeding it beyond rounding error (Supporting Information). Trace preservation is needed: a unital, completely positive generator that creates population gives $\Phi_P > 1$ (Supporting Information).

Competence condition. With $k_f = \frac{1}{4}k_Se^{-\Delta/k_BT}$, eq 4 gives $k_f\Phi_P \leq e^{-\Delta/k_BT}/(4/k_S + 1/k_P)$, so a pair that carries k_{cat} must satisfy

$$k_{\min}(\Delta)\left(\frac{4}{k_S} + \frac{1}{k_P}\right) \leq 1, \qquad k_{\min}(\Delta) \equiv k_{\mathrm{cat}}\, e^{\Delta/k_BT}. \tag{6}$$

Without spin dynamics the flux is exactly $e^{-\Delta/k_BT}/(4/k_S + 4/k_P)$. Singlet–triplet mixing therefore relaxes only the onward term, and by exactly the number of spin states, because triplet pairs cannot return; this is the meaning of the factor four. Eq 6 implies $k_S > 4k_{\min}$ and $k_P > k_{\min}$. It contains no electronic coupling, no reorganization energy and no assumption about the chemistry of the onward step; it is a statement about the generator of eq 2, whose validity for a strongly coupled contact pair is discussed in Sec. VI. If, in addition, no elementary rate exceeds $\nu_n \approx 10^{13}$–10^{14} s^{-1}, at or above the transition-state prefactor $k_BT/h = 6.5\times10^{12}$ s^{-1} at 310 K,[34] eq 6 with $k_S, k_P \leq \nu_n$ requires $k_{\min} \leq \nu_n/5$, that is $\Delta \leq k_BT\ln(\nu_n/5k_{\mathrm{cat}}) = 0.57$–$0.82$ eV for $k_{\mathrm{cat}} = 1$–10^3 s^{-1}. Most of this limit does not need eq 4. $\Phi_P \leq 1$ alone requires $k_f \geq k_{\mathrm{cat}}$, and a pair formed over a barrier no lower than the free energy of its singlet, with a prefactor of at most ν_n, then has $\Delta \leq k_BT\ln(\nu_n/4k_{\mathrm{cat}})$, whatever its spin dynamics and whether or not eq 2 describes it; this is the comparison of activation and electron-transfer free energies used for MAO by Walker and Edmondson,[11] with the spin statistics included. Eq 6 lowers this limit by only $k_BT\ln(5/4) \approx 6$ meV. What it adds is the condition on the onward rate, which the spin dynamics relaxes by at most the factor four.

Dead-end pairs. A radical pair on a branch that does not lead to product carries no flux, so eq 6 does not apply to it, but it holds enzyme, and a field that changes its population changes the free enzyme. With $k_P = 0$ the resolvent need not exist (triplet states that never reach the singlet are stationary), but the identity $k_S\int_0^t e^{\mathscr{L}_0u}(P_S)\,du = I - e^{\mathscr{L}_0t}(I) \preceq I$, with $\mathscr{L}_0$ the generator at $k_P = 0$, bounds the integrated singlet-born population by $4N/k_S$ for all t. The steady-state population of the branch is therefore at most $4k_f/k_S = e^{-\Delta/k_BT}$ per E·S complex, and so is the fraction of enzyme through which a field can act on it.

IV. THE LARGEST FIELD EFFECT AT GIVEN RATES

The exchange coupling of a contact pair is the sum of a reactive contribution from the back electron transfer and a direct contribution. The reactive part is tied to k_S: for non-adiabatic transfer in the classical Marcus–Hush limit,[19]

$$\frac{2\pi J}{k_S} = \frac{1}{2}\,\mathrm{erfi}\left(\frac{\lambda - \Delta_S}{2\sqrt{\lambda k_BT}}\right), \tag{7}$$

with λ the reorganization energy and Δ_S the driving force of the back transfer (J in frequency units, eq 3; in our sign convention $hJ = E_T - E_S$). The ratio vanishes at the activationless point $\lambda = \Delta_S$, where k_S is largest for a given coupling, and grows rapidly with the activation energy. A fast back transfer therefore does not by itself require a large exchange coupling, but it does when the transfer is activated. We set J equal to this reactive part and admit every λ between 0.2 and 2.0 eV for which the back transfer is non-adiabatic in the sense that its golden-rule rate is at most 0.3 times the transition-state rate $\nu_n e^{-(\lambda-\Delta_S)^2/4\lambda k_B T}$ ($\nu_n = 10^{13}$ s^{-1}); a wider range of λ can only enlarge the set of admitted couplings. For a pair at free energy Δ we then seek

$$F^*(k_S, k_P; \Delta) = \sup_{J \text{ admitted},\ \hat{n}_B,\ s\in[0.2,1]} |\mathrm{MFE}_P|, \qquad (8)$$

and write F for the value that a finite search finds; it is not a proven supremum. For each (k_S, k_P) on a grid of 15 by 13 rates the search scans J on a grid that is dense near zero and extends to $\max(4000\ \mathrm{MHz}, 5(k_S + k_P)/2\pi)$, zooms twice around each of the three largest local maxima, and at every J takes the largest response over $s = 0.2$, 0.35, 0.6 and 1 and 25 field directions over a hemisphere. The largest response at each J is stored, so that F for any Δ and any range of λ is the largest of the stored values inside the admitted interval and of the values at its ends, interpolated linearly between stored points; a fixed λ admits a single J. Direct evaluation at the admitted J, over the same s values and directions, for 40 cells and driving forces near the fixed-λ thresholds gives 2×10^{-4} to 1.21 times the interpolated value: the interpolation overestimates where the response is negligible and underestimates it by up to a factor 1.21 elsewhere, and scaling every fixed-λ value by that factor moves the fixed-λ thresholds by at most 0.02 eV, within the grid resolution (Supporting Information). Beyond the scanned range J takes the edge value. With J unconstrained, the same search gives $F_{\mathrm{free}} \geq F$.

Figure 1 shows F_{free} over $k_S = 10^{-2}$–10^5 μs^{-1} and $k_P = 10^{-2}$–10^4 μs^{-1}. The onward rate controls the response: $\max_{k_S} F_{\mathrm{free}}$ is 2.2 % at $k_P = 10^{-2}$ μs^{-1}, 0.011 % at 10^2 μs^{-1} and 1.1×10^{-7} % at 10^4 μs^{-1}. A fast singlet back-transfer does not by itself remove the effect, because the triplet pair cannot return and is drained only by k_P,[18] but a fast spin-independent decay removes it for every k_S, J and s searched.

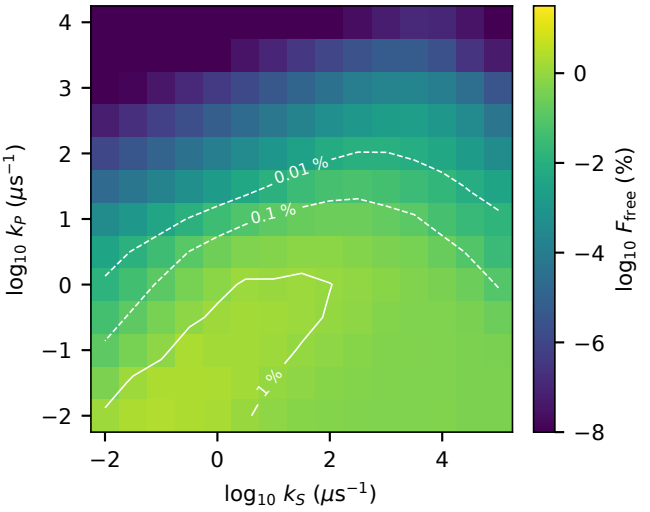


FIG. 1. Largest magnetic field effect on the onward yield found by the search over the exchange coupling (unconstrained), the field direction and the dipolar scale $s \in [0.2, 1]$, as a function of the singlet back-transfer rate k_S and the spin-independent onward rate k_P. Four-nucleus model with anisotropic flavin N5 and N10 tensors (Sec. II C), coherent spin dynamics, $B = 50\,\mu$T. Contours: 1, 0.1 and 0.01 %.

V. FIELD EFFECTS OF A COMPETENT PAIR

Combining eq 6 with the field effect defines

$$G^*(\Delta) = \sup\{\, F^*(k_S, k_P; \Delta) : k_{\min}(\Delta)(4/k_S + 1/k_P) \leq 1 \,\}. \qquad (9)$$

The region in braces is defined by a necessary condition: it contains every pair that can carry the turnover, but not every pair in it can. G^* is therefore an upper bound on the field effect of a competent pair, taken over a larger set. We evaluate it on the rate grid and call the result G: a grid cell stands for the rates up to the next grid rate and is included when its upper corner satisfies the condition. G is a finite-search value at grid rates, not a proven bound. With J tied to Δ through eq 7 the admitted set need not shrink as Δ grows, and we show the running maximum of G from larger Δ. This envelope moves no threshold, because G stays below a level for every larger Δ exactly when its envelope does, and for λ admitted over 0.2–2.0 eV or J unconstrained it changes no value. Where permitted rates lie below the computed grid, or the required rates above it, G is not reported; for $k_{\mathrm{cat}} = 1$ s^{-1} it is defined for $\Delta = 0.25$–0.61 eV.

Figure 2 shows G for $k_{\mathrm{cat}} = 1$–10^3 s^{-1}. Because k_P is sampled every half decade, thresholds are resolved to about 0.03 eV. For $k_{\mathrm{cat}} = 1$ s^{-1}, G falls below 0.1 % at $\Delta = 0.47$ eV and below 0.01 % at 0.52 eV, and stays below up to the largest Δ computed; for $k_{\mathrm{cat}} = 10^3$ s^{-1} the 0.1 % level is reached at 0.28 eV. Fixing λ at 0.5, 1.0 or 1.5 eV instead of admitting the whole range gives 0.1 % thresholds of 0.41–0.47 eV for $k_{\mathrm{cat}} = 1$ s^{-1}, the lowest for $\lambda = 1.5$ eV, and leaving J unconstrained gives the same value, 0.47 eV, because J changes sign at $\lambda = \Delta_S$ and the admitted interval contains the coupling of largest effect. The 0.1 % and 0.01 % levels are reference levels, not detection limits; Fig. 2 gives G over ten decades so that any other level can be read off. MFE_P bounds the relative change of enzyme activity (Sec. II A); the zero-field onward yield of the pair that sets G at the 0.1 % threshold is 0.19, so the absolute change of Φ_P there is about 1×10^{-4} (Supporting Information). A pair near thermoneutrality is not excluded: at small Δ competence permits slow decay and the field effect can be several percent.

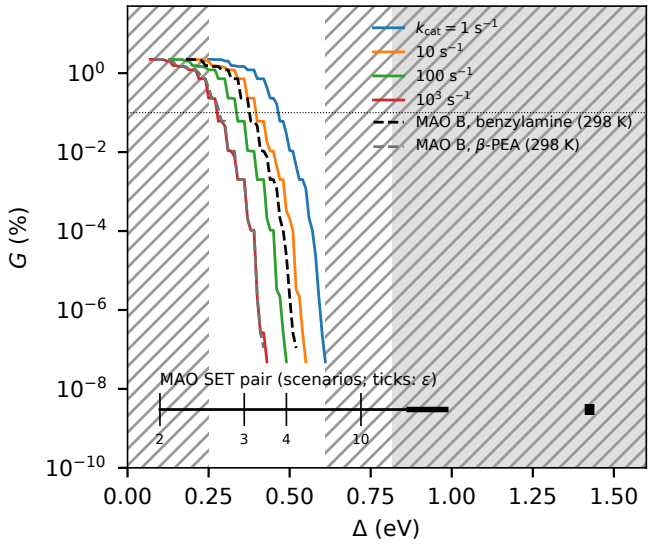


FIG. 2. Largest magnetic field effect found among pairs that satisfy the competence condition, eq 6, $G(\Delta)$, for $k_{cat} = 1$, 10, 100 and 10^3 s^{-1}, with the exchange coupling tied to k_S (eq 7, $\lambda = 0.2$–2.0 eV). Hatched: G not defined for $k_{cat} = 1$ s^{-1} because the required rates lie outside the computed grid. Dashed: the measured rates of flavin reduction of MAO B, 12.7 s^{-1} (benzylamine) and 572 s^{-1} (β-phenylethylamine), at 298 K. Dotted: the 0.1 % reference level. Shaded: no pair carries $k_{cat} = 1$ s^{-1} if no rate exceeds $\nu_n = 10^{14}$ s^{-1}. Bar: illustrative driving forces of the hypothetical SET pair of monoamine oxidase (Sec. VII): scenario (a), thick; (b), medium; (c), thin, with ticks at $\varepsilon = 10$, 4, 3 and 2.

VI. DEPENDENCE ON THE MODEL

Dipolar range and exchange. If the dipolar scale is allowed over $s \in [0,3]$ and J is left free, a full search of the model with isotropic flavin couplings gives a 0.1 % threshold of 0.47 eV for $k_{cat} = 1$ s^{-1}, and the same search with anisotropic flavin tensors on the threshold-setting rows gives 0.50 eV (Supporting Information). Restricting the dipolar scale to $s \geq 0.2$ and tying J to k_S (Sec. IV) gives 0.47 eV for the anisotropic model. The value is set by the assumed largest separation: at the cell that sets it, $(k_S, k_P) = (316, 31.6)$ μs^{-1}, F_{free} is 0.06 % at $s = 0.2$, 0.017 % at $s = 1$ and 0.21 % at $s = 0.01$ (Supporting Information), so pairs farther apart than about 6 Å, or with a dipolar coupling cancelled by exchange, would raise it.

Further nuclei. Adding the aminium ^{14}N or the two largest flavin methyl protons was examined only by searching again around the optimum of the row-maximum cell (Supporting Information). Such a local search can miss larger values elsewhere in the row, and we do not use it to move the thresholds. Several omitted couplings exceed that of N10: the aminium nitrogen (24.8 MHz, spin 1), two further aminium protons (23.0 and 20.7 MHz) and the flavin methyl protons (17.9 MHz; Supporting Information). The thresholds are those of the four-nucleus model.

Electron spin relaxation. We tested one relaxation model, single-electron dephasing with $T_2^e = 10$ ns, 100 ns and 1 μs, in the model with isotropic flavin couplings and J free, on every cell that the thresholds use on the rows $k_P = 10$–$10^{2.5}$ μs^{-1} with $k_S \geq k_P$ (90 searches, with a solver that does not build the Liouvillian; Supporting Information). Dephasing changed individual cells by factors of 0.11–4.94 and the row maxima by factors of 0.84–0.99; with each cell replaced by the larger of its coherent and dephased values, the 0.1 % threshold for $k_{cat} = 1$ s^{-1} is 0.47 eV, unchanged. This statement is restricted to the dephasing model tested; other relaxation mechanisms were not examined.

Validity of the description. The electronic coupling between flavin and substrate does not enter eq 6, but it indicates the regime. Fragment-orbital couplings[35] for methylamine at van der Waals contact (69 non-clashing geometries of 112; Supporting Information) are 53 meV (median) to 106 meV at 3.5 Å, large enough that back electron transfer at contact is likely to be adiabatic rather than non-adiabatic.[36] The search does not tie the dipolar scale s, the electronic coupling implied by k_S and λ, and J to one geometry: the non-adiabatic condition acts on λ, and at $\Delta = 0.5$ eV it leaves admissible values for every one of the 195 cells. The reaction terms of eq 2 are derived for weak, non-adiabatic coupling and were tested against exact results for diabatic couplings of up to 3 meV.[19] Within the search we exclude back transfers that are not non-adiabatic, but for a strongly coupled contact pair the radical-pair and closed-shell states mix, singlet-selective first-order recombination is no longer a clean description, and eq 2, and with it eqs 4 and 6, may not apply. We do not claim the bounds for that regime; they hold for any pair described by eq 2 with admissible relaxation, including the reactive dephasing that appears at higher order in the coupling.

VII. THE HYPOTHETICAL SET PAIR OF MONOAMINE OXIDASE

Rates to be carried. At saturating substrate the flavin of bovine MAO B is reduced with a limiting rate of 760 min^{-1} (12.7 s^{-1}) by benzylamine[11] and of 34 300 min^{-1} (572 s^{-1}) by β-phenylethylamine,[37] at 25 °C; for benzylamine the reduction limits turnover, whereas for β-phenylethylamine it is about 30 times faster than turnover.[37] An SET pair on the reaction path would have to carry these rates per E·S complex; we evaluate them at 298 K. Already $\Phi_P \leq 1$ with a formation prefactor of at most $\nu_n = 10^{13}$–10^{14} s^{-1} (Sec. III) requires $\Delta \leq 0.67$–0.73 eV for benzylamine and $\Delta \leq 0.57$–0.63 eV for β-phenylethylamine, without eq 2; eq 6 lowers these limits by 6 meV.

Thermodynamic cycle. The driving force of the pair in the active site is

$$\Delta = e\left[E^{\circ}_{aq}(\mathrm{RNH_2^{\bullet+}/RNH_2}) + \delta_{am}\right] - e\left[E^{enz}_{ox/sq} + \delta_{sub}\right] + w, \tag{10}$$

where E°_{aq} is the potential of the amine couple in water, δ_{am} the change of that couple on transfer from water into the active site, $E^{enz}_{ox/sq}$ the one-electron potential of the enzyme-bound flavin measured without substrate, δ_{sub} its change on substrate binding, and w the Coulomb interaction of the two radical ions in the site, which neither reference measurement con-

tains. The flavin potential is measured in the enzyme, so its environment is already included and no solvation correction is applied to the flavin; correcting the flavin anion for a low-dielectric site as well would count that environment twice. Only E°_{aq} and $E^{\mathrm{enz}}_{\mathrm{ox/sq}}$ are measured; δ_{am}, δ_{sub} and w are not, and the values below are illustrative scenarios, not bounds.

Measured terms. Chemical reduction of the covalent flavin of MAO proceeds in two steps, the first giving the semiquinone and each requiring two electron equivalents of the enzyme; the oxidized/semiquinone midpoint potential is -159 mV in MAO A and -167 mV in MAO B,[38] and the flavin radical of MAO A is anionic,[39] the radical of eq 1. For the amine, dialkylamine radical cations in water have $E^{\circ} = 1.30 \pm 0.05$ V vs NHE, a value its authors give as a general approximation;[20] we are not aware of a measured potential for a primary alkylamine or benzylamine radical cation in water and use this value as a proxy. Its uncertainty is probably larger than stated: the dialkylamine values come close to the practical limit of cyclic voltammetry in water, follow-up reactions of the radical cations can shift peak potentials negative by up to about 0.2 V, and a computed value of 0.80 V for dimethylamine, much lower than the measured ones, has also been reported.[20]

Scenarios. (a) With $\delta_{\mathrm{am}} = \delta_{\mathrm{sub}} = 0$ and w the contact Coulomb term in water (0.03–0.05 eV for 6–3.5 Å), $\Delta \approx 1.41$–1.44 eV. (b) In the presence of substrates the midpoint potential for full reduction shifts by up to $+0.5$ V and no semiquinone forms.[38] If the absence of semiquinone reflects equilibrium, $E^{\mathrm{enz}}_{\mathrm{ox/sq}} + \delta_{\mathrm{sub}}$ lies at or below the full-reduction midpoint, so $\delta_{\mathrm{sub}} \leq 0.45$ V for MAO A; taking the full $+0.5$ V, deliberately beyond this reading, gives $\Delta \approx 0.86$–0.99 eV, including the uncertainty of the amine proxy. (c) A low-dielectric site changes δ_{am} by Born desolvation of the aminium, with a radius of 2.81 Å calibrated to the measured solvation-energy difference of dimethylamine and its radical cation,[20] and strengthens w. Both terms scale with $1/\varepsilon$, and at contact (3.5 Å) the Coulomb term dominates: relative to the lower end of (b), Δ falls to 0.72, 0.49, 0.36 and 0.10 eV for an effective dielectric constant $\varepsilon = 10$, 4, 3 and 2. The flavin keeps its measured environment, so this correction is one-sided. Part of the electrostatic effect of the bound substrate may already be contained in δ_{sub}, so combining (b) and (c) may overstate the lowering, and a point-charge continuum is least reliable at contact.

Result. In scenarios (a) and (b) the pair lies above both limits, 0.67–0.73 eV for benzylamine and 0.57–0.63 eV for β-phenylethylamine, and cannot carry the measured rate of flavin reduction. This limit does not depend on eq 2, the spin model or the search; it restates, with the spin statistics, the argument by which Walker and Edmondson judged the SET mechanism improbable.[11] In scenario (c) the result depends on ε. The largest field effect found exceeds 0.1 % only below $\Delta = 0.38$ eV for the benzylamine rate and 0.29 eV for the β-phenylethylamine rate, which (c) reaches for ε below 3.1 and 2.6. For $\varepsilon = 4$, $\Delta \approx 0.49$ eV and G is 2.4×10^{-5} % for the benzylamine rate; for $\varepsilon = 3$, $\Delta \approx 0.36$ eV and G is 0.23 %; for $\varepsilon = 2$, $\Delta \approx 0.10$ eV and G is 2.2 % for the β-phenylethylamine rate, while for the benzylamine rate competence permits rates below the computed grid. A field effect above 0.1 % in this enzyme would therefore require an active site that lowers the driving force of the contact pair by about 0.5 eV beyond scenario (b), with an effective dielectric constant near 3 or below, where the continuum estimate is least reliable. It is also the route in which the description used here is least secure: a pair at contact has electronic couplings of 53 meV (median) to 106 meV (Sec. VI), so its back transfer is likely adiabatic, and neither eq 2 nor the field effects computed from it are guaranteed to apply. The calculation does not exclude such a site; independent evidence favours the polar mechanism, from structural, kinetic and computational studies,[12–15] from the absence of a detectable flavin radical during reduction of MAO B by benzylamine analogues,[11] and from the absence of field effects on the kinetics of other flavoenzymes.[7] The analysis concerns substrates oxidized at the amine nitrogen with the dialkylamine potential as a proxy; the catechol and indole neurotransmitters would require their own potentials.

VIII. A LIGHT-DRIVEN PAIR IN THE SAME MODEL

The flavin–tryptophan pair of cryptochrome is formed by light,[3,4] so eq 6 does not apply to it. As a check that the same equations produce a field response when the rates are not constrained, we evaluated it with the same scheme and yield observable, averaged over field orientations rather than maximized, $k_S = k_P = 1\ \mu\mathrm{s}^{-1}$, the pair 1.90 nm apart ($D = -11.4$ MHz), flavin N5 and N10 with their isotropic couplings, the two largest β-proton couplings (60 and 56 MHz) and N1 of the tryptophan radical cation from the same density-functional method, and the exchange coupling on 21 values spanning the literature range of its prefactor with both signs.[17] The orientation-averaged response ranges from below 10^{-2} % (it changes sign within the range) to 0.63 %. This is a numerical control, not a like-for-like comparison: the hyperfine couplings, dipolar coupling and rates differ from the enzyme model as well as the way the pair is formed.

IX. DISCUSSION

The yield bound, eq 4, and the competence condition, eq 6, apply to a radical pair that is formed thermally from a closed-shell state, returns to it only through the singlet, carries the catalytic flux through a spin-independent onward step, and is described by eq 2 with trace-preserving, unital, positivity-preserving relaxation; a pair on a dead-end branch is limited instead by the occupancy bound of Sec. III. They do not require any electronic coupling, reorganization energy or particular onward chemistry, and they should not be generalized to radical pairs outside this scope without further analysis. The numbers that follow depend on more: F and G are results of a finite search in a specific spin model, with the exchange coupling tied to the back-transfer rate by eq 7, a dipolar range for stacked radicals and one relaxation model tested; and the driving force of the MAO pair rests on a proxy amine potential and on terms of the thermodynamic cycle that are not measured.

Two routes around the bound should be named. Three-radical and scavenging mechanisms[40] restore sensitivity to strongly coupled pairs. Whether eq 6 limits them depends on where the scavenging step enters the reaction scheme, and their field effects would have to be computed in their own model. And an active site that brings Δ close to zero is not excluded: competence then allows slow decay and field effects of a percent or more. For the MAO pair this would need an effective dielectric constant near 3 or below at contact (Sec. VII). The question for a particular enzyme is therefore where its radical pair, if there is one, lies in Δ, and how fast it must turn over; both are measurable.

X. CONCLUSIONS

For a radical pair that an enzyme forms thermally and through which it turns over, the onward yield obeys $\Phi_P \leq 4k_P/(k_S + 4k_P)$ for any spin Hamiltonian and any trace-preserving, unital, positivity-preserving relaxation, and detailed balance then requires $k_{\mathrm{cat}} e^{\Delta/k_B T}(4/k_S + 1/k_P) \leq 1$. Spin dynamics relaxes the corresponding spin-free condition only in its onward term, by the number of spin states. In a flavin–aminium spin model with anisotropic hyperfine tensors and radicals stacked 3.5–6 Å apart, the largest field effect found among pairs that satisfy this condition for $k_{\mathrm{cat}} = 1\ \mathrm{s}^{-1}$ falls below 0.1 % at $\Delta = 0.47$ eV. Applied to the hypothetical single-electron-transfer pair of monoamine oxidase, with the measured rates of flavin reduction and a thermodynamic cycle built on the measured flavin potentials and a proxy amine potential, the analysis finds that such a pair could not carry the observed flux unless the active site lowered its driving force below about 0.73 eV, and that a field effect above 0.1 % would require an effective dielectric constant near 3 or below at contact.

SUPPLEMENTARY MATERIAL

See the supplementary material for the 300-system test of eq 4, the full search results, the electron-spin-relaxation calculations in the four-nucleus model, the hyperfine and dipolar tensors, the coupling scan, the density-functional driving forces, the model-sensitivity calculations, the cryptochrome scan, and software and environment.

ACKNOWLEDGMENTS

We thank an anonymous reviewer for suggesting the bound of eq 4 and the competence condition of eq 6.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Hikaru Wakaura: Conceptualization (lead); Methodology (lead); Software (lead); Validation (lead); Formal analysis (lead); Investigation (lead); Data curation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Taiki Tanimae:** Supervision (lead).

DATA AVAILABILITY

The data that support the findings of this study and the code that generates them are openly available in the `qbscreen` package at `https://github.com/deeptell-inc/brain_protein_screening`; every number in the text is generated from the stored results by `qbscreen/make_numbers_v6.py`.

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